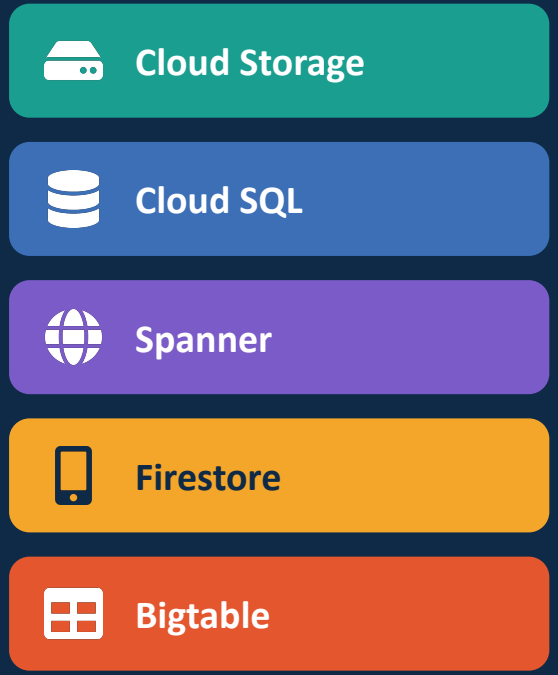
MODULE 04

Storage in the Cloud

Choosing the right home for every kind of data: objects, relational tables, documents and wide-column big data



What we will cover today

01

Cloud Storage

Object storage for files, media, backups and archives

02

Cloud SQL

Managed MySQL, PostgreSQL and SQL Server

03

Spanner

Relational and horizontally scalable to global size

04

Firestore

Document database for mobile and web apps

05

Bigtable

Wide-column NoSQL for huge, fast workloads

06

Comparing the options

Picking the right service for a workload

By the end of this module you will be able to

- Explain object, file and block storage
- Use buckets, storage classes, versioning and lifecycle rules
- Describe when Cloud SQL and Spanner fit relational workloads
- Explain Firestore's document model and offline sync
- Identify workloads that suit Bigtable
- Choose the right storage service for a scenario



Module format

6 topics
12-question knowledge check
1 hands-on lab

Different data needs different storage



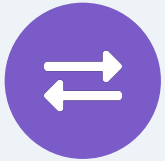
Unstructured

Images, video, audio, logs, backups



Structured

Rows and columns with a fixed schema



Transactional

Orders, payments, bookings that must be consistent



Relational

Tables linked by keys and queried with SQL joins



Ask two questions about any data: what shape is it, and how will it be read and written? The answers point to the service.

01

Cloud Storage



Durable, highly available object storage for any amount of data

Three ways to store data



File storage

Folders and files in a hierarchy, like a shared drive

Example: Filestore



Block storage

Raw chunks of a disk that an operating system formats and uses

Example: Persistent Disk



Object storage

Each item is an object: the data, its metadata and a unique ID, reachable by URL

Example: Cloud Storage

Because every object has a URL, object storage works naturally with the web. Typical objects: video, pictures and audio recordings.

What teams use Cloud Storage for



Website content

Images, scripts and media served to users



Backup, archive, DR

Long-term copies and disaster recovery



Direct download

Distributing large files to end users



Intermediate results

Data passed between steps of a processing pipeline



BLOB storage

A fully managed, scalable home for binary large objects: no servers to run, no capacity to plan.

Buckets hold objects, and objects don't change

Every bucket needs

A globally unique name

No other bucket anywhere in Google Cloud can share it

A location

A region, dual-region or multi-region close to your users, to reduce latency

Objects are immutable



You never edit an object in place. Each change uploads a new version of it.

Example: most users are in Europe → choose a European region or the EU multi-region.

Versioning and lifecycle rules

Object versioning

Off (default)

A new upload overwrites the old object. The old data is gone.

On

Old versions are kept as a history of overwrites and deletes.

With versioning on, you can list old versions, restore one, or permanently delete one.

Lifecycle rules save money

- Delete objects older than 365 days
- Delete objects created before a set date
- Keep only the 3 newest versions of each object
- Move older objects to a colder storage class

Controlling who can access objects

01



IAM roles

- Enough for most needs
- Inherited from project to bucket to object

02



Access control lists

For finer control. Each ACL entry pairs:

Scope	Permission
Who: a user or group	What: such as read or write

Goal: least privilege. Objects can hold personal data, so give each person only the access their job needs.

Four storage classes, from hot to cold



Standard

Frequently accessed ("hot") data, or data kept only briefly

Min: None



Nearline

Read or changed about once a month or less: backups, long-tail media

Min: 30 days



Coldline

Read or changed at most once every 90 days

Min: 90 days



Archive

Accessed less than once a year: archives, backup, disaster recovery

Min: 365 days



Higher storage cost, lower access cost

Lower storage cost, higher access cost

Same experience in every class, plus Autoclass

WHAT EVERY CLASS SHARES

- Unlimited storage, no minimum object size
- Worldwide access and locations
- Low latency and high durability
- Same security, tools and APIs
- Geo-redundancy in dual- and multi-regions

Autoclass

Moves each object to the right class automatically, based on how it's used.

Not accessed

→ moves to colder classes to cut storage cost

Accessed again

→ moves back to Standard for faster future access

Paying, protecting and loading data

Pay for what you use

No minimum fee, no capacity to provision in advance

Encrypted at rest

Server-side encryption before data is written to disk, at no extra cost

Encrypted in transit

HTTPS/TLS between your devices and Google

GETTING DATA IN: PICK BY SIZE

Online transfer

Everyday uploads

gcloud storage commands, or drag and drop in the console

Storage Transfer Service

TB to PB, online

Scheduled batch transfers from other clouds, other regions or HTTP(S) endpoints

Transfer Appliance

Up to 1 PB per appliance

A rackable server you lease, fill with data and ship to Google

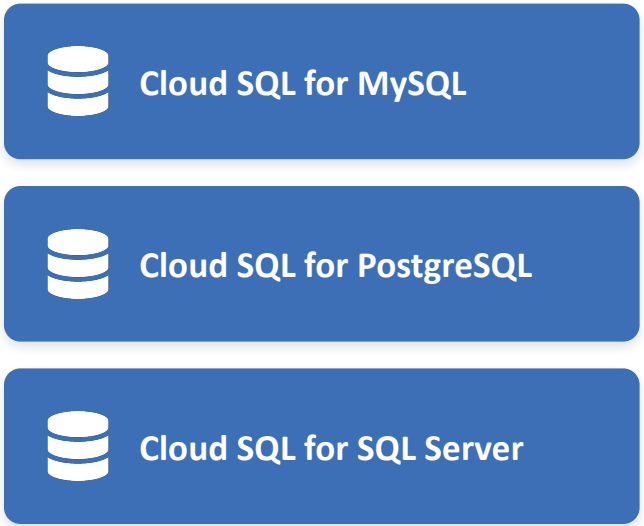
02

Cloud SQL



Fully managed relational databases, so your team focuses on the application, not the database server

Cloud SQL: familiar engines, fully managed



GOOGLE HANDLES THE CHORES



You focus on

Schemas, queries and building the application, instead of installing and maintaining database software.

Why teams choose Cloud SQL



Nothing to install

No database software to install or maintain



Scales up

Up to 128 processor cores, 864 GB RAM and 64 TB of storage



Replication built in

Supports automatic replication scenarios



Managed backups

Instance cost covers seven backups



Encryption

On Google's internal networks and in tables, temp files and backups



Network firewall

Controls network access to each instance

Reachable from App Engine (standard drivers such as Connector/J or MySQLdb), authorized Compute Engine VMs, and tools like SQL Workbench and Toad.

03

Spanner



A relational database that scales horizontally, stays strongly consistent and speaks SQL

Spanner: relational, at global scale



Scales horizontally

Add capacity by adding nodes, not by buying a bigger server



Strongly consistent

Every read sees the latest committed write



Speaks SQL

Familiar relational model and queries

Battle tested

Powers Google's own mission-critical applications and services.

ESPECIALLY SUITED FOR APPLICATIONS THAT NEED

- **SQL with joins and secondary indexes**
- **Built-in high availability**
- **External consistency**
- **Tens of thousands of reads/writes per second**

Cloud SQL or Spanner?

	Cloud SQL	Spanner
Data model	Relational, SQL	Relational, SQL
How it scales	Vertically (bigger machine) plus read replicas	Horizontally, by adding nodes
Capacity	Up to 64 TB	Petabytes
Best for	Web frameworks and existing apps: user accounts, orders	Global, high-throughput transactional systems
Engines	MySQL, PostgreSQL, SQL Server	Spanner (GoogleSQL or PostgreSQL dialect)

Rule of thumb: start with Cloud SQL. Move to Spanner when you need horizontal scale, not just read replicas.

04

Firestore



A flexible, horizontally scalable NoSQL document database for mobile, web and server apps

Firestore stores documents in collections



Collection

`users`

Groups related documents





Document: alovelace

```
first : "Ada"

last : "Lovelace"

born : 1815

address : { city: "London" }
```

Key-value pairs, including nested objects

+ subcollections

- Incoming data is stored as documents
- Documents are organized into collections
- Documents can hold nested objects and subcollections
- No fixed schema: documents can differ

Queries, offline sync and Google-scale reliability

QUERIES

- Fetch one document or all that match
- Chain multiple filters
- Combine filtering and sorting
- Indexed by default: speed scales with result size, not dataset size



Works offline

Caches the data an app is using, so it can write, read, listen and query with no connection.

Back online → local changes sync to Firestore automatically.

GOOGLE-SCALE INFRASTRUCTURE

- ✓ Automatic multi-region replication
- ✓ Strong consistency guarantees
- ✓ Atomic batch operations
- ✓ Real transaction support

05

Bigtable



Google's NoSQL big data database, for massive workloads at consistently low latency

Bigtable: massive scale, consistently fast



Massive workloads



Consistent low latency



High throughput

The same database behind Google services

Search, Analytics, Maps and Gmail all run on Bigtable.

Great for operational and analytical apps: IoT, user analytics, financial data analysis.

CUSTOMERS CHOOSE BIGTABLE WHEN

1. More than 1 TB of structured or semi-structured data
2. Fast, high-throughput or rapidly changing data
3. NoSQL data, without strong relational needs
4. Time-series or naturally ordered data
5. Big data batch or real-time processing
6. Machine learning on the data

Getting data in and out of Bigtable

APIs

Serving apps and dashboards

Through a data service layer such as an HBase REST Server or a Java server using the HBase client

Streaming

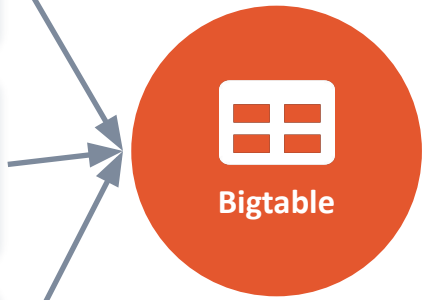
Real-time ingestion

Stream-processing frameworks such as Dataflow Streaming, Spark Streaming and Storm

Batch

Periodic large jobs

Hadoop MapReduce, Dataflow or Spark; results are often written back to Bigtable



Not the same as BigQuery: BigQuery is a data warehouse for SQL analytics, sitting between storage and processing.

06

Comparing storage options

Matching each workload to the service that fits it best

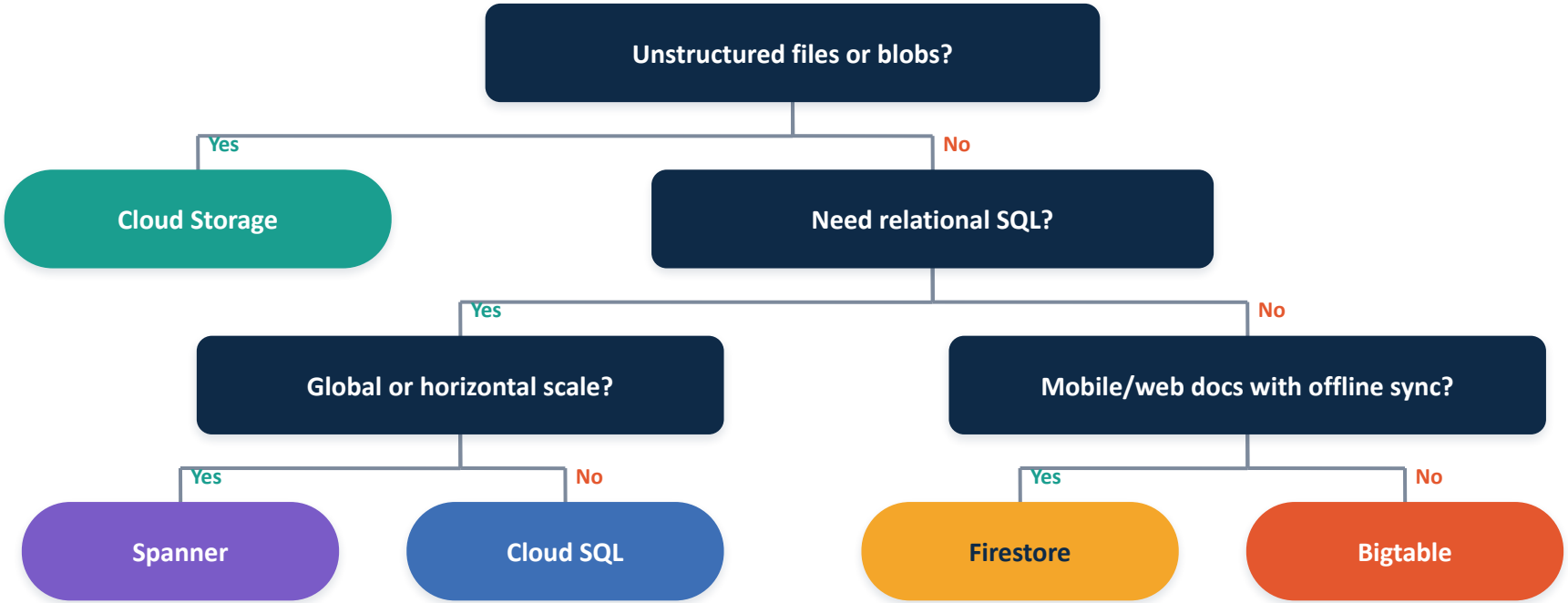


The five services side by side

Service	Type	Best for	Capacity
Cloud Storage	Object	Immutable blobs larger than 10 MB, such as large images and movies	Petabytes; 5 TB max per object
Cloud SQL	Relational	Full SQL for OLTP; web frameworks and existing apps	Up to 64 TB
Spanner	Relational	Full SQL for OLTP with horizontal scalability	Petabytes
Firestore	NoSQL document	Massive scale with real-time and offline queries for mobile and web apps	Terabytes; 1 MB max per entity
Bigtable	NoSQL wide-column	Large volumes of structured objects with heavy reads and writes; no SQL or multi-row transactions	Petabytes; 10 MB per cell, 100 MB per row

Real applications often combine several of these services.

A quick way to choose



A starting point, not a rule. Check volume, latency and consistency needs too. (No branch: huge analytical or time-series data → Bigtable.)

Match the workload to the service

SCENARIO	BEST FIT
A media site stores 4 GB video files for download	**Cloud Storage**
An existing WordPress site needs a managed MySQL database	**Cloud SQL**
A global bank needs consistent transactions across continents	**Spanner**
A to-do app must keep working offline on phones	**Firestore**
Millions of IoT sensor readings arrive every second	**Bigtable**

Key takeaways

1. **Cloud Storage**
   Object storage in buckets; pick a class by access frequency

2. **Cost control**
   Versioning, lifecycle rules and Autoclass

3. **Cloud SQL**
   Managed MySQL, PostgreSQL and SQL Server up to 64 TB

4. **Spanner**
   Relational and strongly consistent, scaling horizontally

5. **Firestore**
   Document NoSQL with real-time and offline sync

6. **Bigtable**
   Wide-column NoSQL for massive, fast, time-series data

?

Knowledge check



12 questions covering all six topics. Answer each one before we reveal it.

Question 1 of 6

Storage types

Which storage model keeps each item as data plus metadata plus a unique ID that can be reached by URL?

A File storage

B Block storage

C Object storage

D Relational storage

Hint (if needed): Think of how Cloud Storage serves an image on a website.

Answer 1: C

Storage types

Which storage model keeps each item as data plus metadata plus a unique ID that can be reached by URL?

A File storage

B Block storage

✓ **Object storage**

D Relational storage

Why

Object storage packages the data, its metadata and a globally unique identifier, usually a URL. Cloud Storage is Google's object storage service.

Question 2 of 6

Buckets

What must be true of every Cloud Storage bucket name?

A. It must match the project name

B. It must be unique within the project

C. It must be globally unique across Google Cloud

D. It must include the region name



Hint (if needed): Think about why the lab named the bucket after the project ID.

Answer 2: C

Buckets

What must be true of every Cloud Storage bucket name?

A It must match the project name

B It must be unique within the project

It must be globally unique across Google Cloud

D It must include the region name

Why

Bucket names share a single global namespace, so each name must be unique across all of Google Cloud.

Question 3 of 6

Versioning

Versioning is enabled on a bucket. A user uploads a new copy of report.pdf. What happens to the previous copy?

A. It's overwritten and lost

B. It's kept as an older version that can be restored

C. The upload is rejected because objects are immutable

D. It moves automatically to Archive storage



Hint (if needed): That's the purpose of versioning.

Answer 3: B

Versioning

Versioning is enabled on a bucket. A user uploads a new copy of report.pdf. What happens to the previous copy?

A It's overwritten and lost

✓ **It's kept as an older version that can be restored**

C The upload is rejected because objects are immutable

D It moves automatically to Archive storage

Why

With versioning on, Cloud Storage keeps a history of overwrites and deletes. You can list, restore or permanently delete older versions.

Question 4 of 6

Lifecycle

Which feature can automatically delete objects older than 365 days?

A Object versioning

B Object Lifecycle Management

C Access control lists

D Storage Transfer Service



Hint (if needed): These are rules applied to a bucket.

Answer 4: B

Lifecycle

Which feature can automatically delete objects older than 365 days?

A Object versioning

✓ **Object Lifecycle Management**

C Access control lists

D Storage Transfer Service

✓ **Why**

Lifecycle management policies act on objects by age, creation date or number of versions, for example deleting objects older than 365 days.

Question 5 of 6

Storage classes

Monthly reports are read about once a month and kept for several years. Which storage class fits best?

A Standard

B Nearline

C Coldline

D Archive

Hint (if needed): Match the class to the access frequency.

Answer 5: B

Storage classes

Monthly reports are read about once a month and kept for several years. Which storage class fits best?

A Standard

✓ **Nearline**

C Coldline

D Archive

Why

Nearline is designed for data read or modified about once a month or less, such as backups and long-tail content.

Question 6 of 6

Storage classes

Which storage class has the lowest storage cost and a 365-day minimum storage duration?

A Nearline

B Coldline

C Standard

D Archive



Hint (if needed): It's the coldest class.

Answer 6: D

Storage classes

Which storage class has the lowest storage cost and a 365-day minimum storage duration?

A Nearline

B Coldline

C Standard

✓ **Archive**

Why

Archive is the lowest-cost class, intended for data accessed less than once a year, with a 365-day minimum storage duration.

Up next: hands-on lab



Getting Started with Cloud Storage and Cloud SQL

Build a small PHP web page on a Compute Engine VM that reads from Cloud SQL and shows an image stored in Cloud Storage.

1. Launch the bloghost web server VM
2. Create a bucket and upload a public image
3. Create a Cloud SQL instance and authorize the VM
4. Connect the page to the database and add the image

THANK YOU

Questions and discussion



Cloud Storage | Cloud SQL | Spanner | Firestore | Bigtable | Choosing the right service